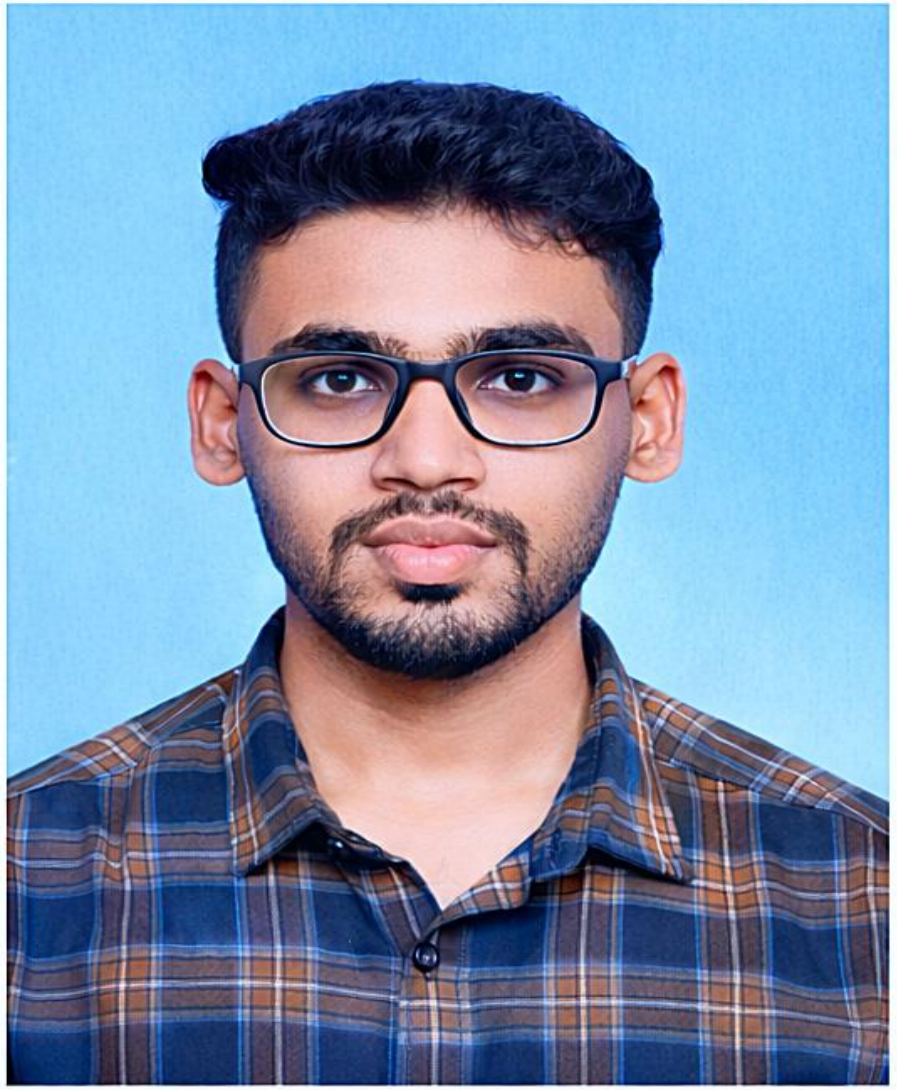
Microsoft Elevate

Project Title:

Consumer Financial Complaints Analysis



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Project Background

Financial institutions receive thousands of consumer complaints related to financial products such as credit cards, loans, and debt collection.

The **Consumer Financial Protection Bureau (CFPB)** collects and publishes these complaints to monitor consumer protection risks.

Analyzing this dataset helps identify:

- Major drivers of consumer complaints
- Financial products generating the most dissatisfaction
- Effectiveness of company complaint resolution
- Patterns in consumer disputes

Problem Statement

The increasing volume of complaints makes it difficult for regulators and stakeholders to quickly identify major consumer issues.

Key questions addressed in this project:

- Which financial products generate the highest complaints?
- Which companies receive the most complaints?
- What issues are most frequently reported by consumers?
- How effective are company responses in resolving complaints?
- Which complaints result in consumer disputes?

Data Preparation

The CFPB complaints dataset was processed using **PostgreSQL** before visualization.

Key preparation steps:

- Imported dataset into PostgreSQL database
- Filtered data for complaints from **2020–2025**
- Removed records with missing critical fields
- Standardized product categories
- Extracted time features (year, month)
- Created analytical views for reporting
- This ensured the dataset was **clean, structured, and ready for analysis**.

Complaint Overview Dashboard

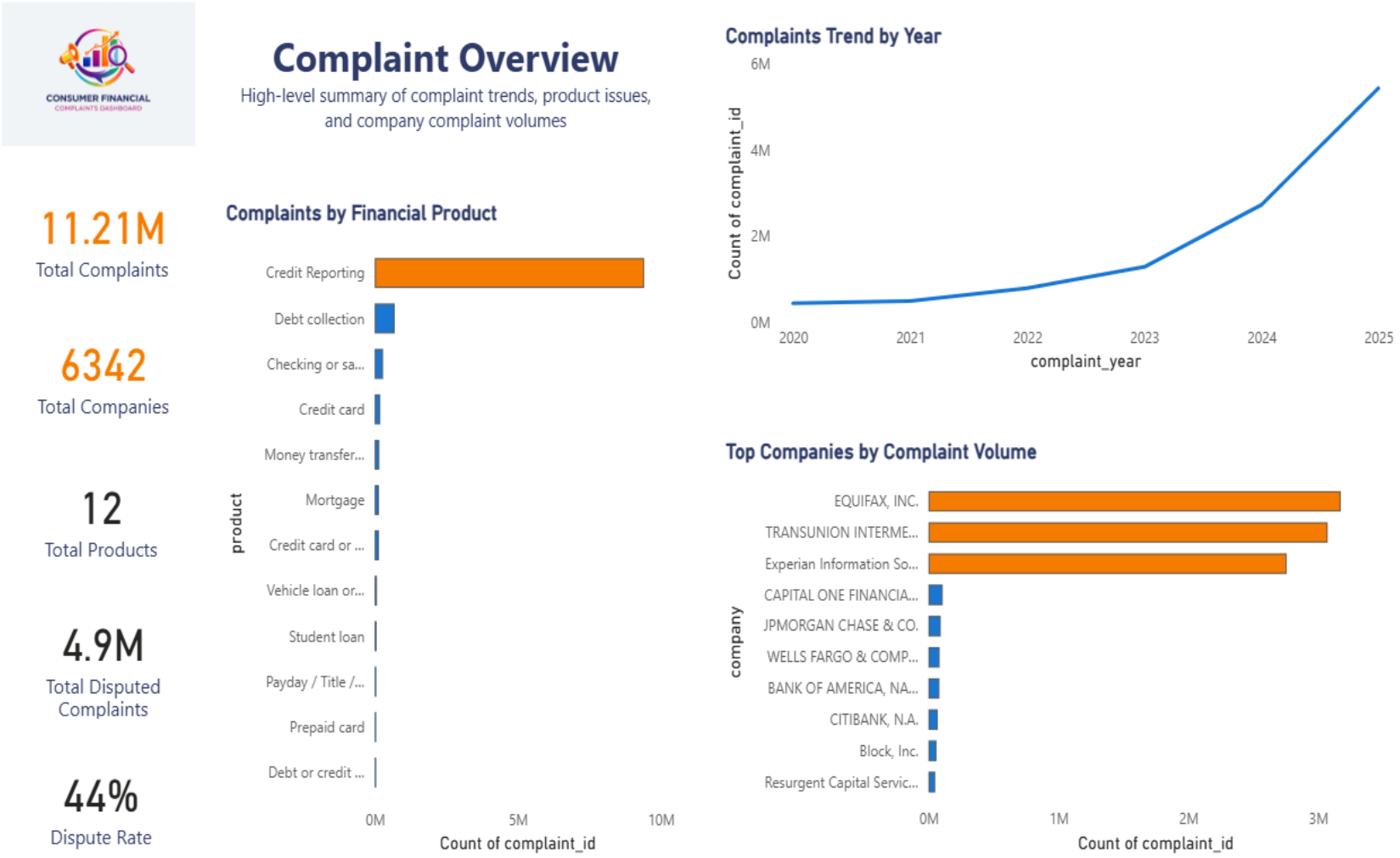
Key Dashboard Insights:

This dashboard provides a high-level overview of consumer financial complaints reported to the CFPB.

Key insights include:

- Total number of complaints received between 2020–2025
- Complaint trends over time to identify growth in consumer issues
- Financial products generating the highest complaint volumes
- Companies receiving the most consumer complaints

This overview helps stakeholders quickly identify **major complaint drivers and high-risk financial products**.



Complaint Issue Analysis Dashboard

Key Dashboard Insights:

This dashboard analyzes the types of problems consumers report when filing complaints.

The analysis highlights:

- Most frequently reported consumer complaint issues

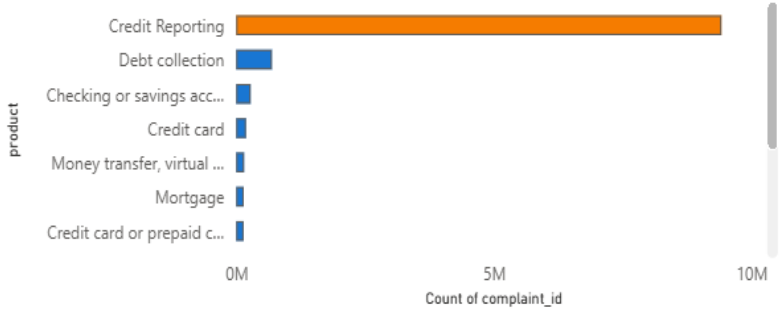
- Complaint distribution across financial products and sub-products

- Relationships between financial products and specific complaint issues

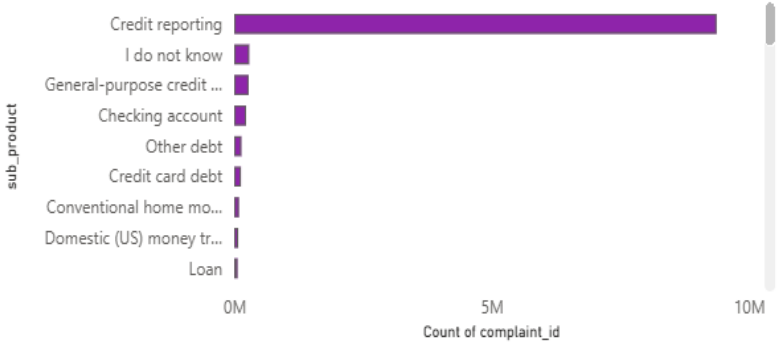
This helps identify **systemic problems in financial services such as inaccurate credit reporting and debt collection issues**.

Complaint Issue Analysis

Complaints by Financial Product



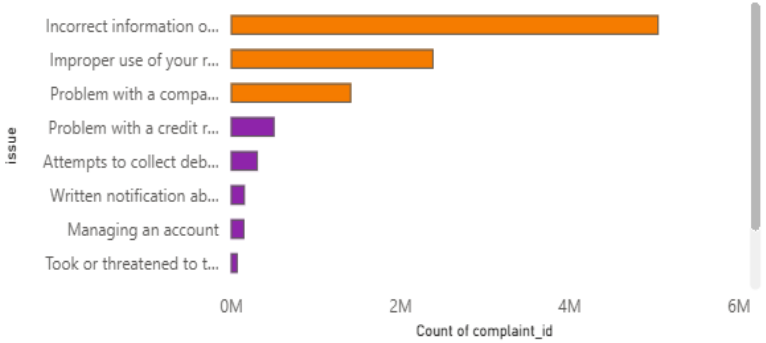
Complaints by Financial Sub-Product



Product vs Issue Breakdown

product	Attempts to collect debt not owed	False statements or representation	Improper use of your report
Checking or savings account			235
Credit card			1896
Credit card or prepaid card			1109
Credit Reporting			2389383
Debt collection	320143	75652	
Mortgage			565
Payday / Title / Personal Loan			750
Student loan			1842
Vehicle loan or lease			1240
Total	**320143**	**75652**	**2397020**

Most Reported Consumer Issues



Dispute & Resolution Analysis Dashboard

Key Dashboard Insights:

This dashboard evaluates how effectively companies resolve consumer complaints.

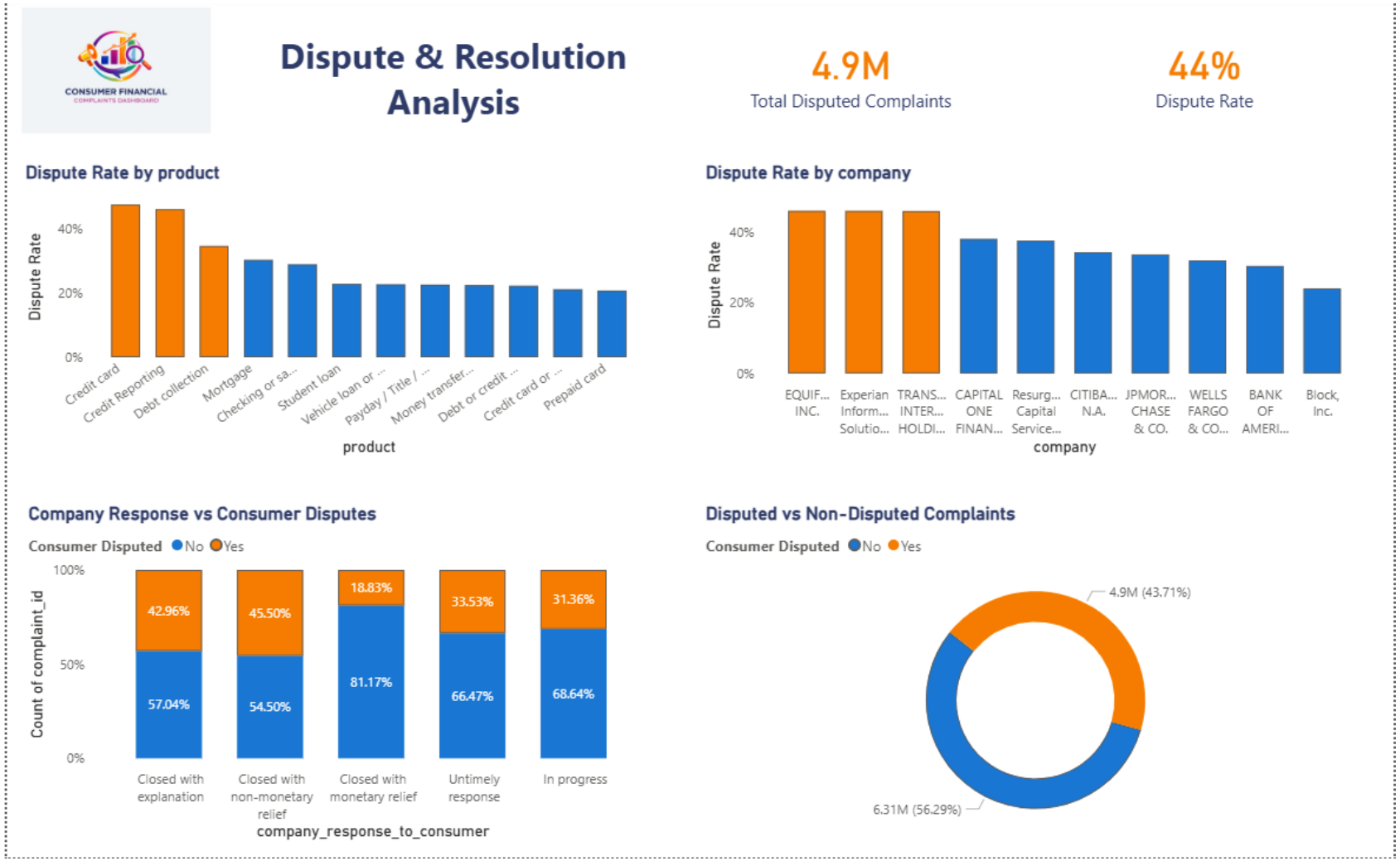
Key insights include:

- Dispute rates across financial products and companies

- Overall proportion of complaints that remain disputed

- Impact of company response types on dispute outcomes

The analysis shows that complaints resolved with **monetary relief have significantly lower dispute rates**, indicating more effective complaint resolution.



Key Insights:

Credit reporting and credit cards generate the highest complaint volumes, indicating major consumer concerns in these product categories.

The most frequently reported issue involves **incorrect information on credit reports**, suggesting data accuracy challenges.

Certain companies receive significantly higher complaint volumes, highlighting areas requiring regulatory attention.

Approximately **44% of complaints remain disputed**, indicating that many company responses do not fully resolve consumer concerns.

Complaints resolved with **monetary relief show the lowest dispute rates**, suggesting financial compensation is the most effective resolution strategy.

Recommendations:

Based on the analysis, the following actions are recommended:

1. Improve complaint resolution processes

2. Increase transparency in credit reporting systems

3. Strengthen oversight of debt collection practices

4. Encourage companies to prioritize **effective resolutions over explanatory responses**

Using complaint analytics regularly can help identify **emerging risks and improve consumer protection**.

Conclusion:

This project demonstrates how complaint data can be used to identify systemic issues in financial services.

By analyzing complaint patterns, organizations can:

- Detect high-risk financial products
- Evaluate company response effectiveness
- Improve consumer complaint resolution strategies

The Power BI dashboard enables **data-driven monitoring of consumer dissatisfaction and complaint trends**.

REFERENCES:

Consumer Financial Protection Bureau (CFPB)

Dataset Source:
https://www.consumerfinance.gov/data-research/consumer-complaints/

Tools Used

PostgreSQL – Data cleaning and preparation
Power BI Desktop – Visualization and insights
Power BI Service – Dashboard sharing

GitHub Link:
https://github.com/VarunJoshi-Max/Consumer-Financial-Complaints-Analysis.git

Power BI Service Dashboard Link:

https://app.powerbi.com/links/f-TQpYn-tw?ctid=5cf28a99-8d4b-415c-a59e-cd6a57d65be6&pbi_source=linkShare

Thank You